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INFRASTRUCTURE SWEDEN

SciLifeLab

Image registration for microscopy images

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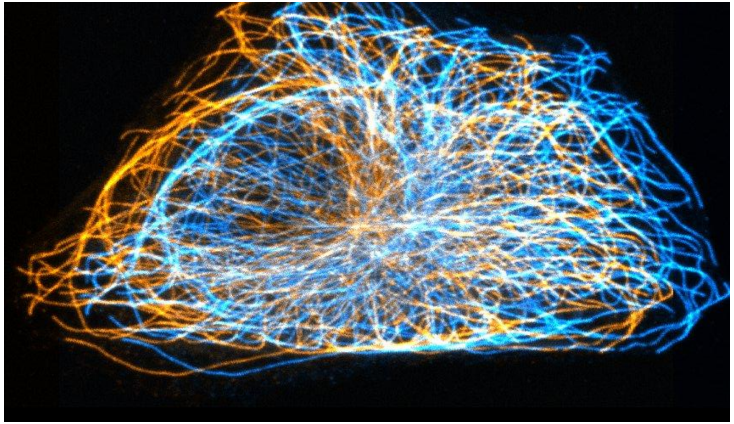
Outline

- What is image registration?
- Methods in image registration
- Types of transformations
- Available tools
- Demonstration

Image registration

- Image registration also termed as image alignment is the process of aligning (overlaying) different images over each other
- It could also be done on different sets of image data to bring to one coordinate system
- It is necessary in order to be able to compare or integrate the data obtained from different measurements



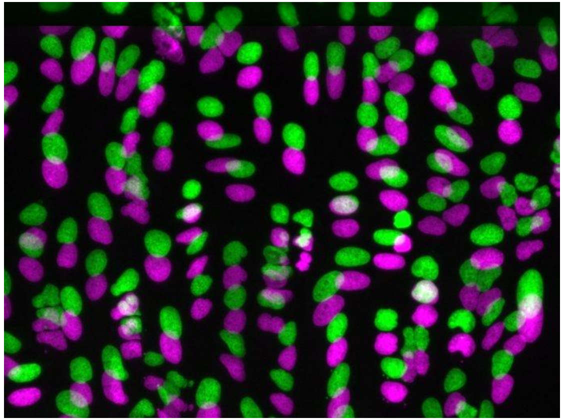
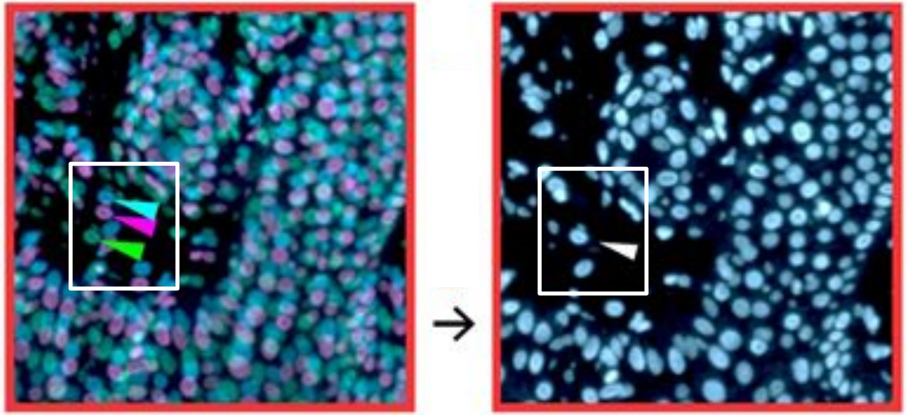


Image from BBBC (Gustafsdottir *et al.*, 2013)

Scenarios

Cyclic imaging



Tiled images

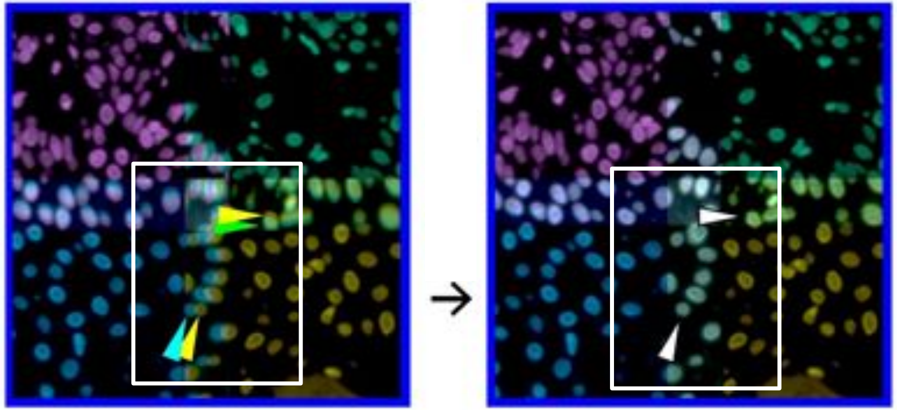
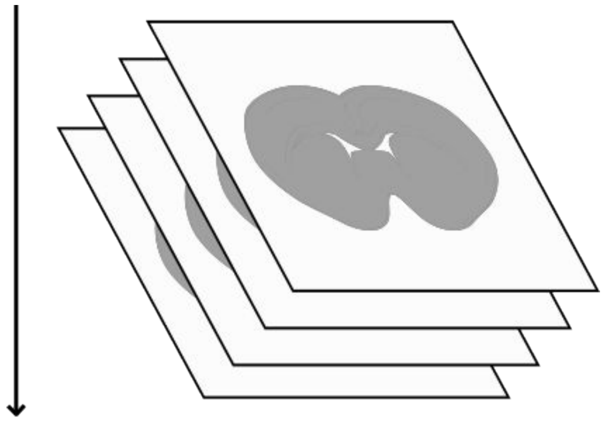


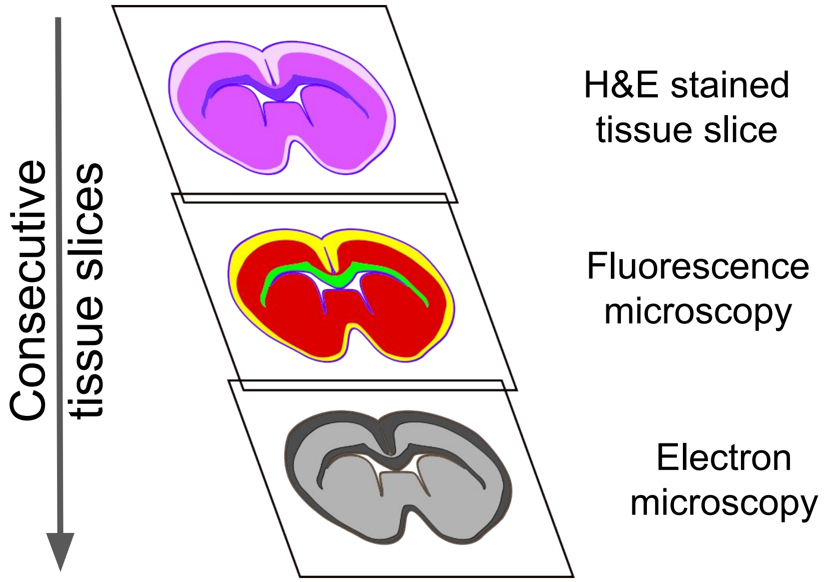
Image adapted from Muhlich *et al* (2022), Bioinformatics under [CC BY license]()

Scenarios

Serial sections



Multimodal images



Basic terminology

Input image / ***Moving image*** - the image that will be transformed

Reference image / Fixed image - is the image against which the input image is registered

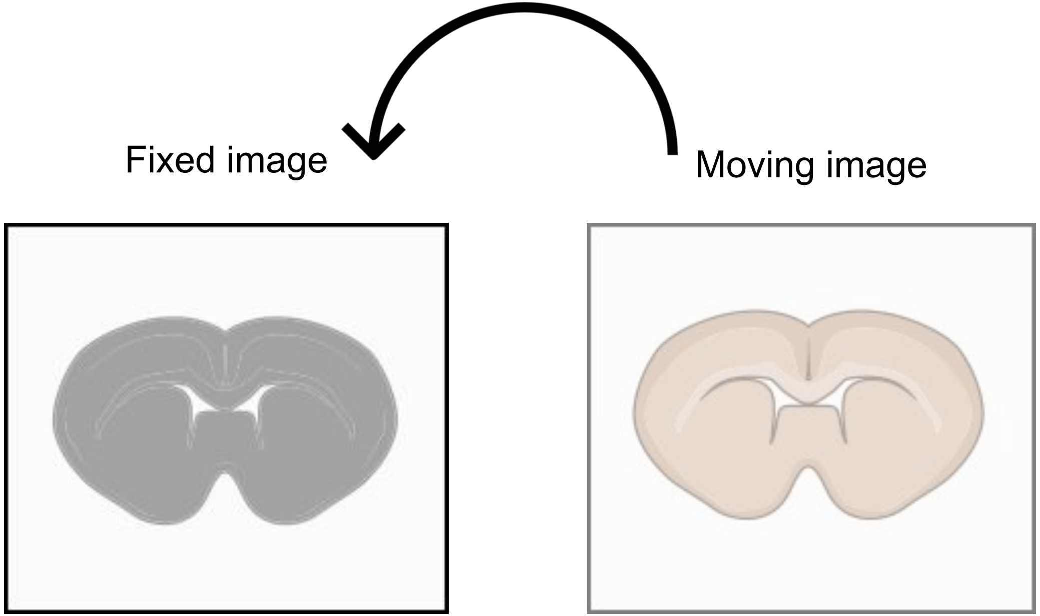
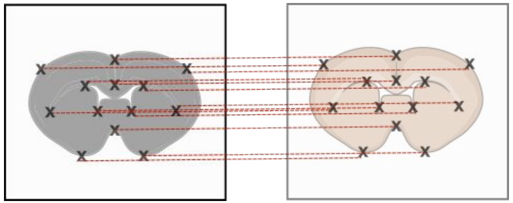


Image registration - Methods

Image Registration

Feature based



Intensity based

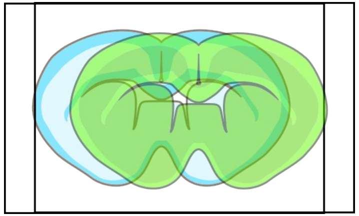
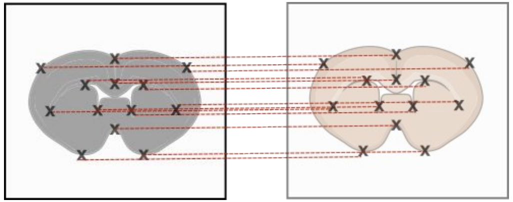


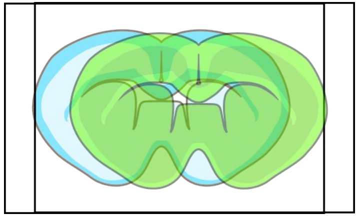
Image registration - Methods

Image Registration

Feature based

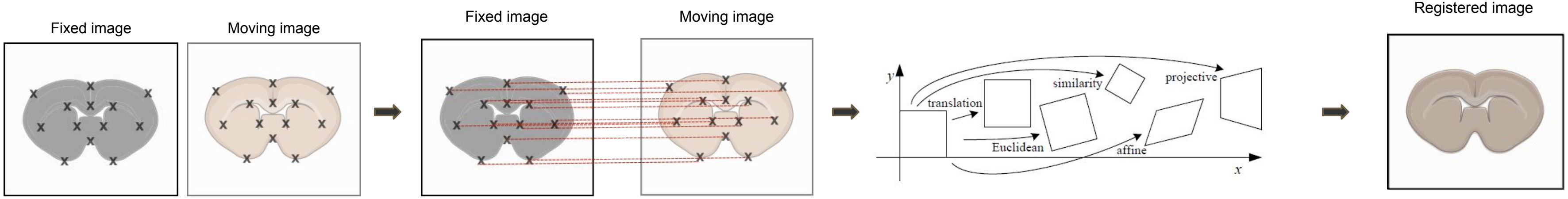


Intensity based



Feature-based image registration

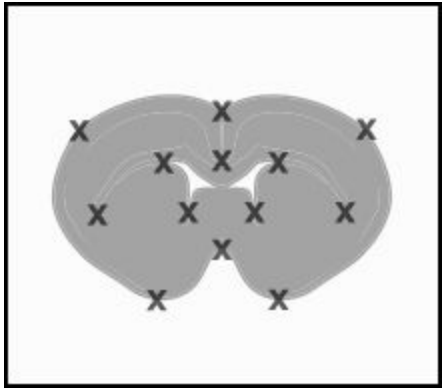
Feature detection → Feature matching → Transform model estimation → Image transformation



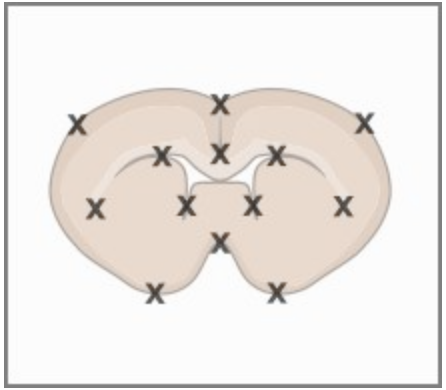
Feature detection

- Salient and distinctive objects that are common in both the images are considered
- Point features, line features, edges, region features are some examples
- These are called ***control points/ keypoints / correspondence points***
- The detected points should be spread over the images, stable in time and not be sensitive to illumination changes.

Fixed image

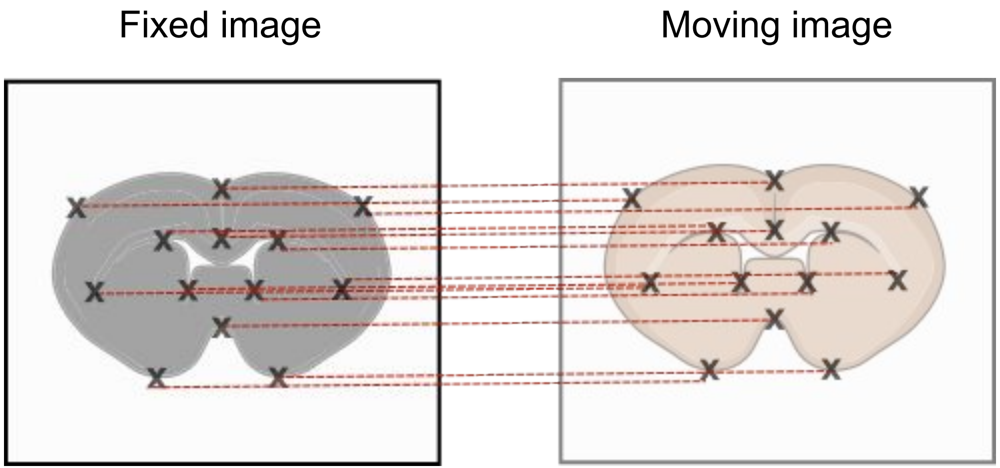


Moving image



Feature matching

- The correspondence between the features detected in the input image and those detected in the reference image is established.
- Feature descriptors and similarity measures along with spatial relationships among the features are used for matching.



Transform model estimation - Methods

- Linear / non-deformable transformation
- Non-linear / deformable transformation

Linear transformations

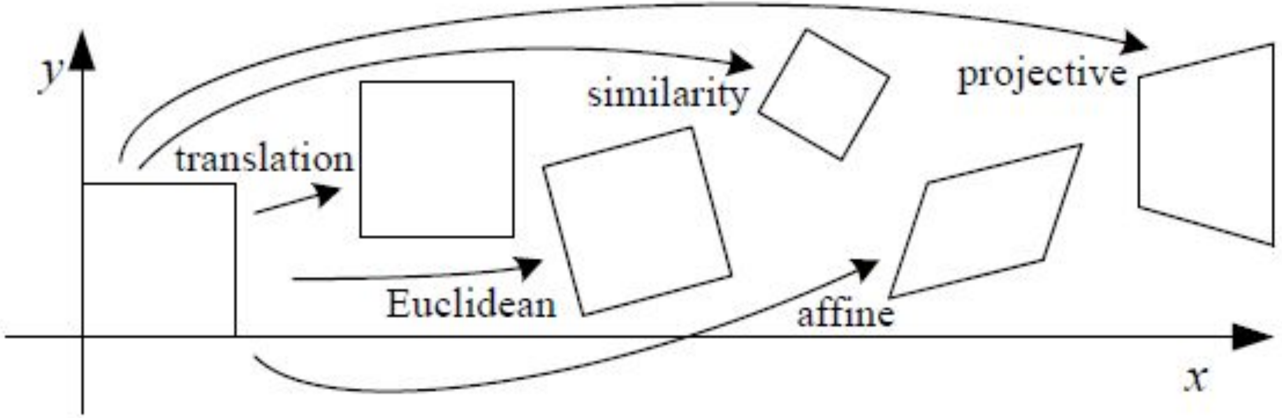
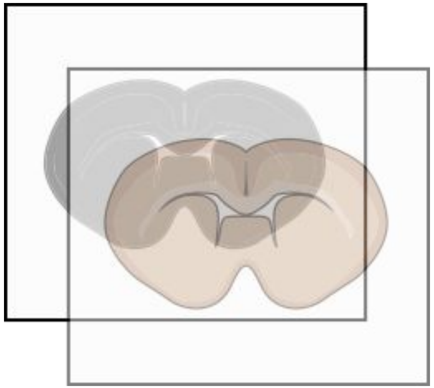
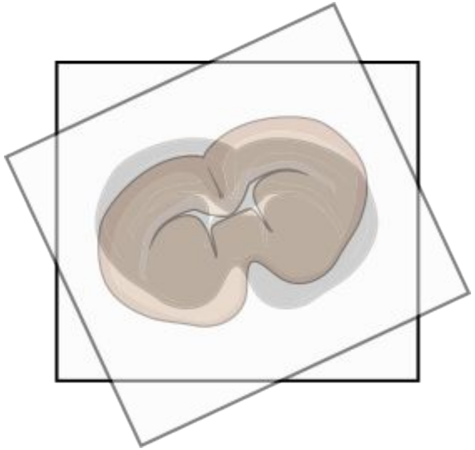


Image from Szeliski, R. (2022). *Computer Vision: Algorithms and Applications*, 2nd ed. Springer.

Linear transformations

Rigid/Euclidean transformation



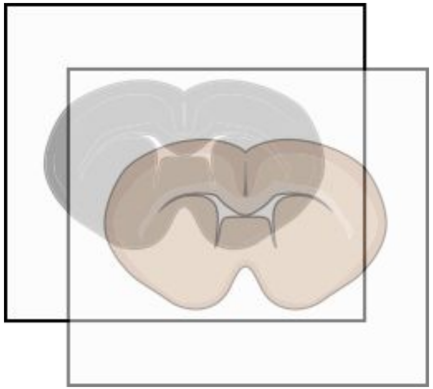


Translation - parameters - 2 (x, y) ; preserves orientation
Rigid transformation - parameters - 3 (x, y, ϴ); preserves lengths

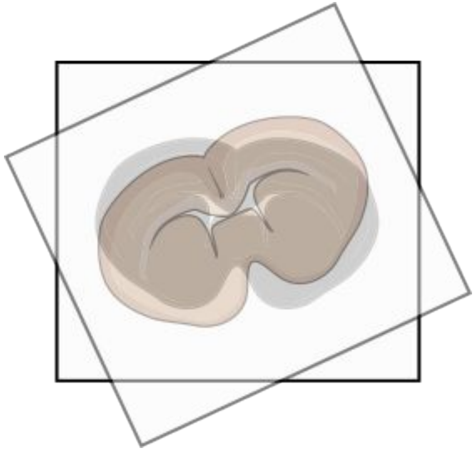
Linear transformations

Similarity transformation

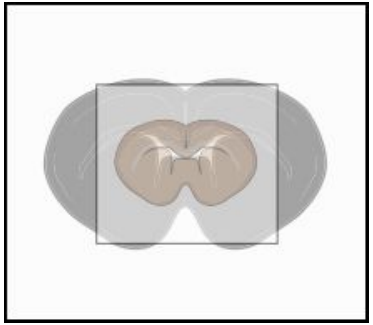
Translation



Rotation



Scaling

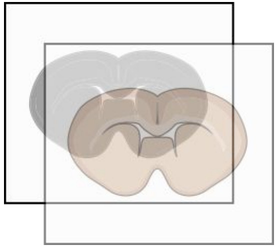


Similarity transformation - parameters - 4; preserves angles, parallelism and straight lines

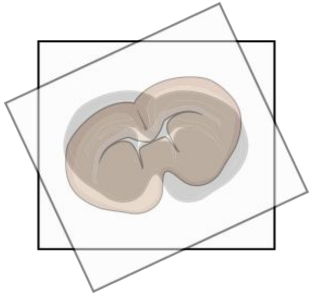
Linear transformations

Affine transformation

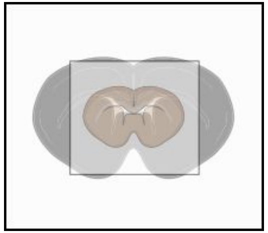
Translation



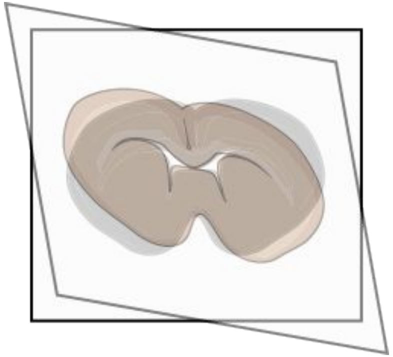
Rotation



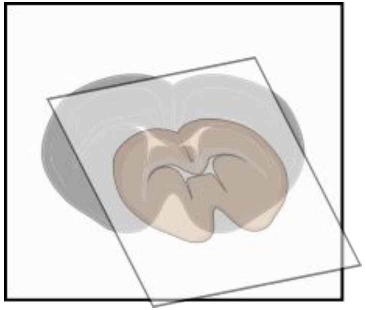
Scaling



Shear



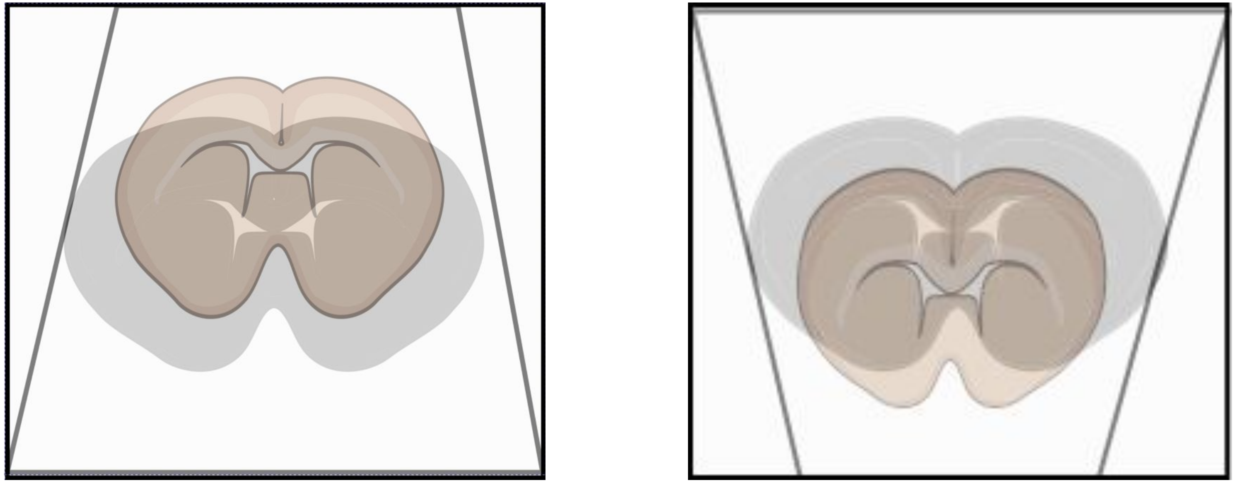
Example of Affine



Affine transformations - parameters - 6; preserves parallelism

Linear transformations

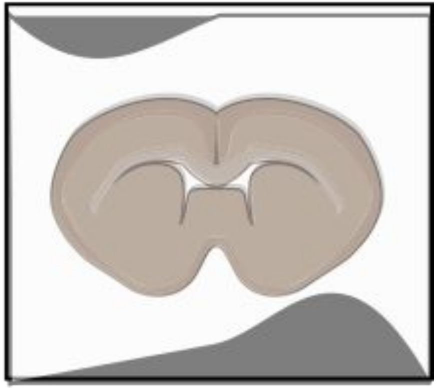
Projective transformation

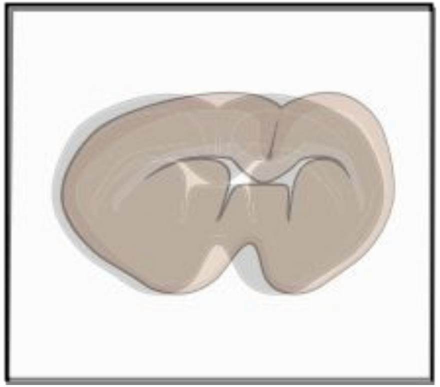


Projective transformations - parameters - 8; preserves straight lines

Non-linear transformations

Deformable transformation



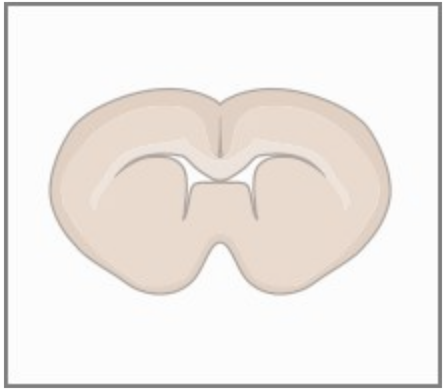


Large set of parameters

How do the transformations take place?

Linear transformations

Moving image



Scaling & Rotation

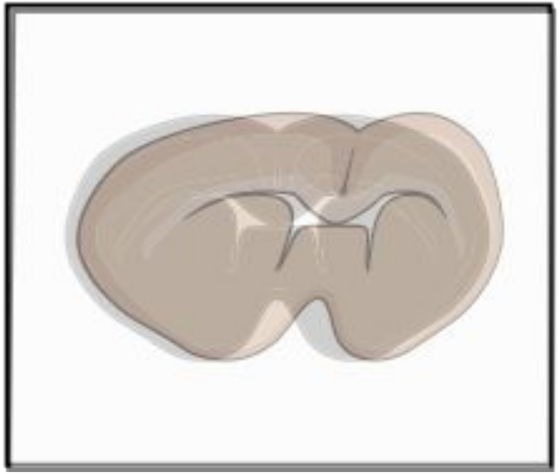
Translation

$$\times \left\{ \begin{array}{ccc} a_{11} & a_{12} & t_1 \\ a_{21} & a_{22} & t_2 \\ 0 & 0 & 1 \end{array} \right\}$$

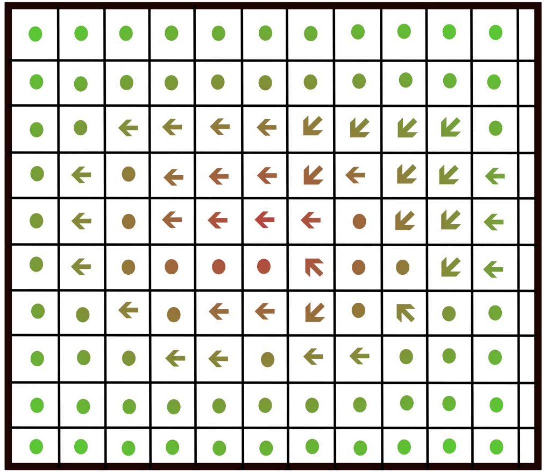
How do the transformations take place?

Non-linear transformations

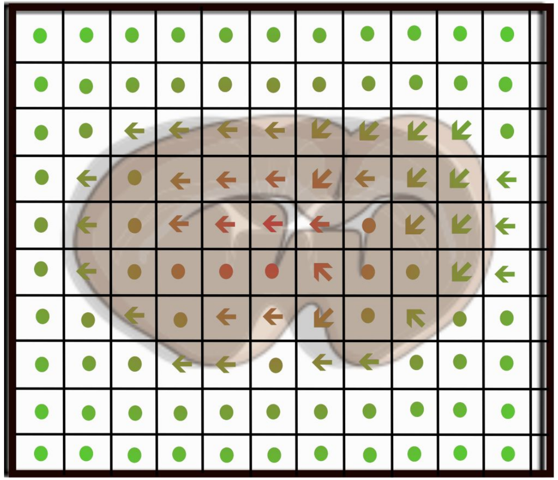
Sample image



Displacement



Overlay



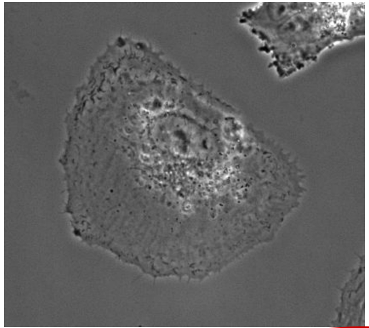
Evaluation

Pearson correlation coefficient - higher the better

Sum of squared differences - lower the better

Example

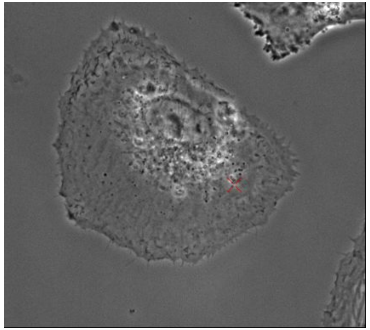
Frame 1



Pixel values

Prefs	137	138	139	140	141	142	143
63	$\boxed{103}$	105	97	97	97	95	103
64	95	103	99	109	104	103	108
65	78	84	98	110	101	110	104
66	77	80	95	113	113	105	109
67	70	66	86	109	111	108	105
68	65	71	75	102	116	117	121
69	59	72	79	103	127	138	133

Frame 2



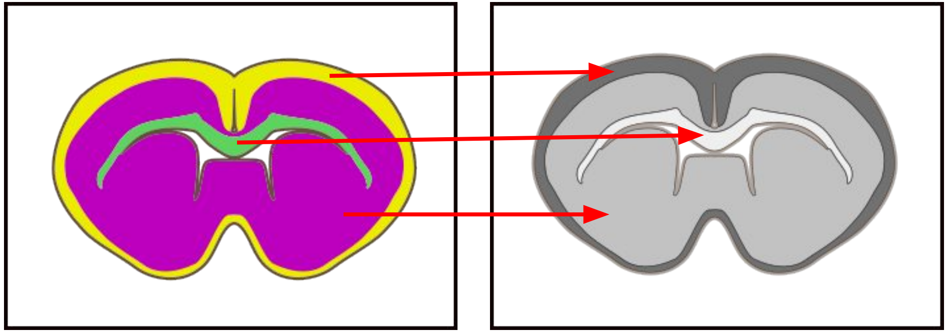
Pixel values

Prefs	137	138	139	140	141	142	143
63	$\boxed{93}$	97	106	91	71	80	90
64	89	90	81	56	63	82	90
65	88	70	45	52	81	91	90
66	80	54	51	68	90	95	88
67	81	73	78	91	102	99	106
68	81	92	98	99	103	109	102
69	95	96	100	109	101	112	103

Evaluation

Mutual information - it is a measure of co-occurrence of intensity between the two images

Fixed image | Moving image



Moving image (columns) / Fixed image (rows)

	0	0	200
	0	300	0
	100	0	0

Evaluation

Increasing mutual information value

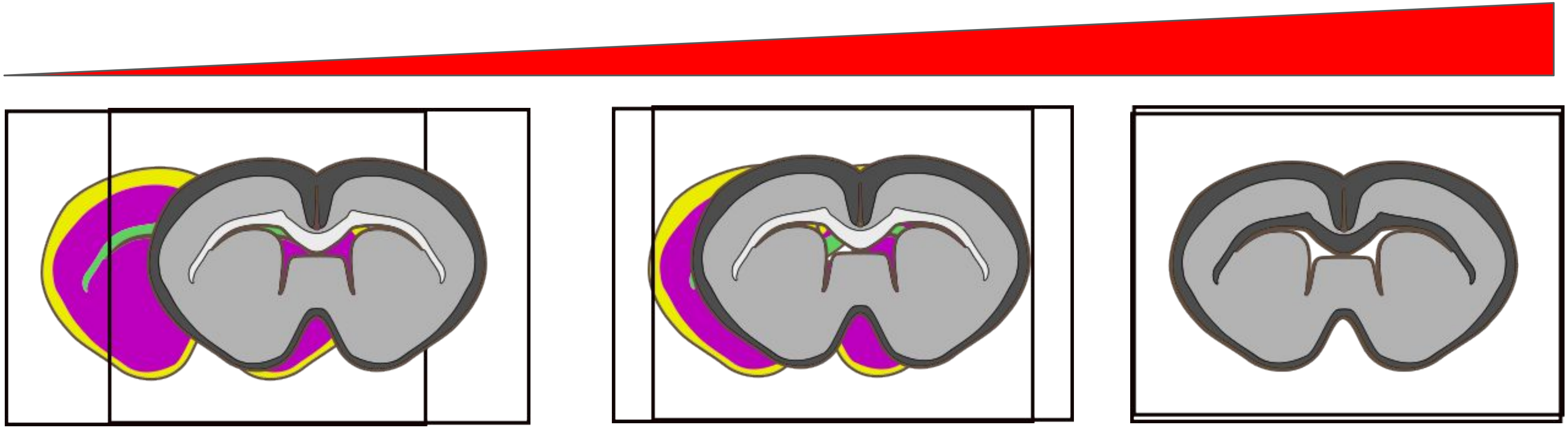
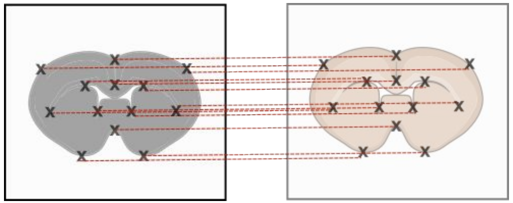


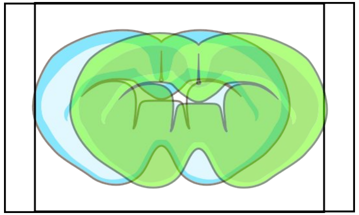
Image registration - Methods

Image Registration

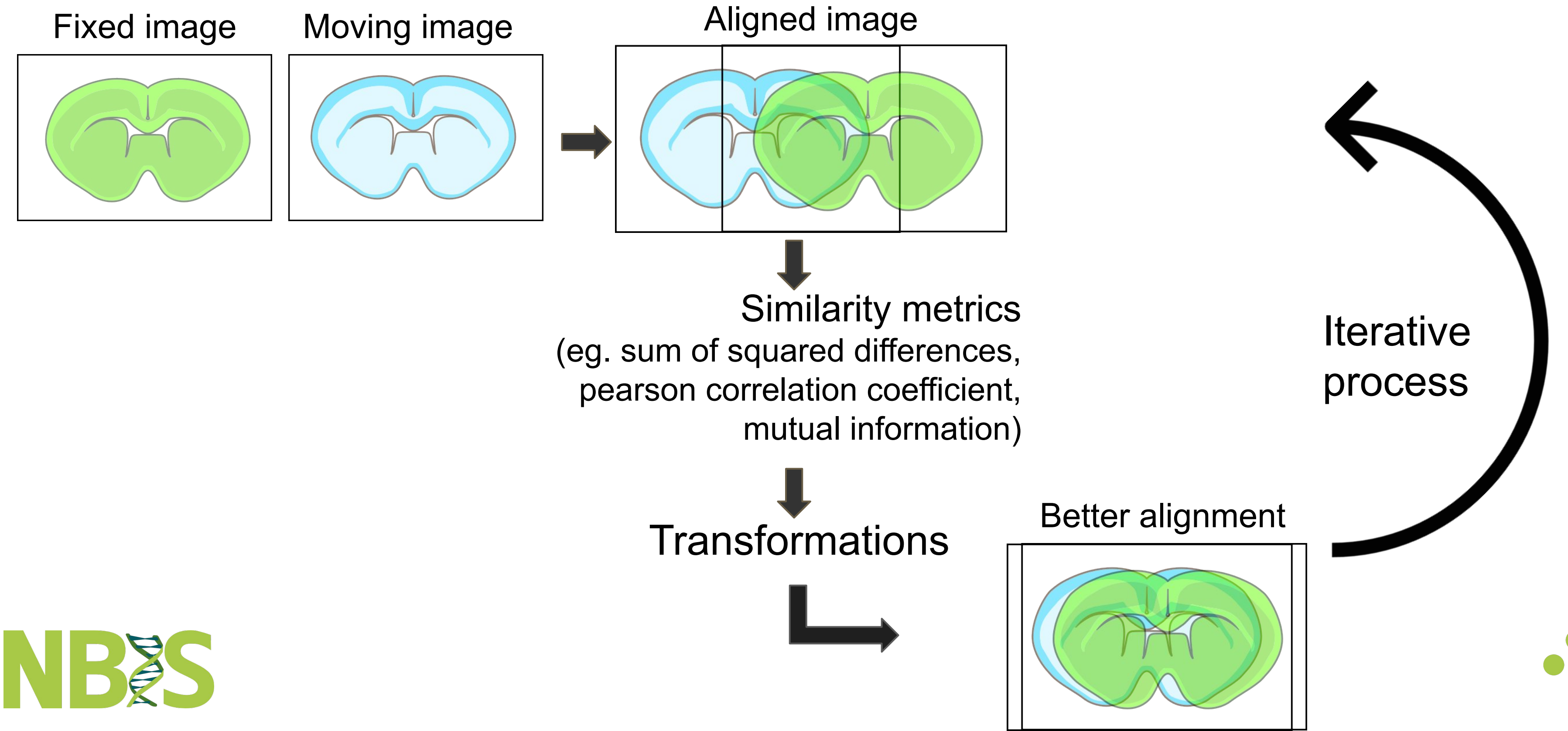
Feature based



Intensity based



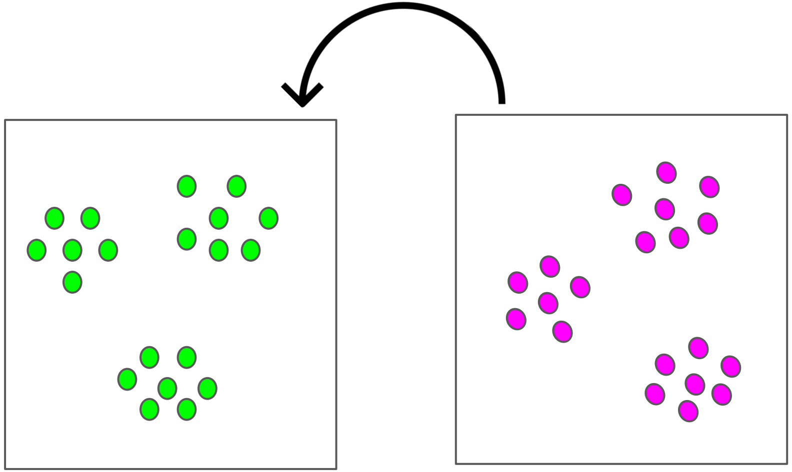
Intensity-based registration



Point set registration

- Assigning correspondence between two sets of points and to determine the transformation that maps one point set to the other

- Methods - Coherent Point Drift (CPD) algorithm, Iterative Closest Point (ICP) and many more



Tools

Tool	Usage	Method of transformation
Warpy	Manual	Used for more complex transformations than affine. Uses ImLib2 library, elastix and BigWarp
BigWarp	Manual	Landmark-based, Transformations - rigid, affine, similarity, thin plate spline
Wsireg	Semi-automated	Uses elastix for transformation Supports linear and non-linear transformation models
Fast4DReg	Automated	A Fiji macro. Mostly used for drift correction but can also be used for aligning multi-channel 2D or 3D images
MultiStackReg	Automated	A Fiji plugin; Based on TurboReg. Mostly for aligning stacks
VALIS	Automated	Rigid and non-rigid transformations
ANTsPy	Automated	Rigid and deformable transformations

Current developments

- ITK-Wasm - image registrations and spatial analysis can be carried out in a web browser
- Information about the transformations and the coordinate systems can be made available (soon) in the OME-Zarr format. For more details, check out here

Resources

- ITK-Elastix - GitHub
- Elastix documentation
- For more image registration tools - BioImage Informatics Index
- If you have any questions, feel free to post it on image sc forum

References

- Jeremy L Muhlich, Yu-An Chen, Clarence Yapp, Douglas Russell, Sandro Santagata, Peter K Sorger, Stitching and registering highly multiplexed whole-slide images of tissues and tumors using ASHLAR, *Bioinformatics*, Volume 38, Issue 19, October 2022, Pages 4613–4621, https://doi.org/10.1093/bioinformatics/btac544
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- Blampey, Q., Mulder, K., Gardet, M. *et al.* Sopa: a technology-invariant pipeline for analyses of image-based spatial omics. *Nat Commun* 15, 4981 (2024). https://doi.org/10.1038/s41467-024-48981-z
- All the mouse brain images used for illustration were from NIAID Visual & Medical Arts. (10/7/2024). Mouse Brain Coronal. NIAID NIH BIOART Source. bioart.niaid.nih.gov/bioart/370
- Some images used for illustration (as indicated) come from the Broad Bioimage Benchmark Collection specifically the dataset from Gustafsdottir et al., 2013
- Images that have no indications are from Bhuvanasundar Renganathan, Research Associate, Northwestern University, Chicago

BioImage Informatics Unit (BIIF)

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Acknowledgement

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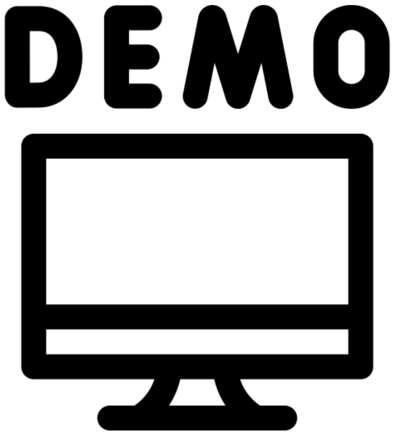
- ***Natasa Sladoje***, Professor, Dept. of Information Technology, Uppsala University, Uppsala
- ***Bhuvanasundar Renganathan***, Research Associate, Northwestern University, Chicago

Questions

Demonstration



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